

Personal Objects as Design Materials:

A Case Study of Co-Designing Safe Spaces With Young Adults

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ABSTRACT

Design materials are often imposed on users in co-design. This can be problematic as the materials may not be relevant or useful to users' contexts or goals, or, at worst, could lead to alienation and exclusion. We propose an approach that shifts control of design materials to users, supporting them to select and utilize personal objects in the design process. We conducted a case study alongside a STEM education nonprofit based in Chicago to explore how 9 young adults brought in and engaged with personal objects in the co-design of a "Safe Spaces" event for peers. We found that the personal objects helped inform participants' designs, structure the conversation, and foster inclusion and common ground. We provide suggestions for how to support users bringing in and using personal objects as design materials.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods; Participatory design.

KEYWORDS

Participatory Design, Co-Design, Design Materials, Materiality, Inclusion

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1 INTRODUCTION

Participatory design strives to be inclusive of users and their lived experiences, but may sometimes fall short of this in practice [14, 24, 25]. Despite being an integral component of the design process [34] and human experience [7, 13, 38], the selection and use of design materials is not only overlooked, but can also adversely affect or exacerbate this exclusivity. For example, design materials (e.g., sticky notes [23], crayons [31], microcontrollers, tablets) are often ascribed to users despite not being relevant or useful to users' goals

or contexts. These materials are often value-laden and overlook users' situated contexts [4, 20]. Users may also prefer to draw on their own tangible or intangible artifacts, such as oral narratives [4], metaphors and symbols [1], and skills and wisdom [40]. *What if, instead of facilitators imposing materials on users, we shift control by inviting users to bring in objects from their lives to use as design materials? What are some mechanisms to support this process, as well as the myriad of possible (tangible and intangible) objects users may want to bring in?*

In this paper, we present preliminary findings from a case study with a STEM (science, technology, engineering, math) education nonprofit in Chicago focused on providing out-of-school opportunities for underrepresented young adults. We collaborated with nonprofit leadership to embed our innovative concept and approach in a co-design workshop focused on creating a "Safe Spaces" kickback event by and for young adults. We observed how 9 young adults (ages 16–21) brought in and engaged with a myriad of personal objects in the design of the kickback event for peers. We found that personal objects and our "4 C's" approach helped inform participants' event design, add structure to the workshop, and connect participants to one another and foster common ground. We contribute an empirical investigation into how users can leverage personal objects in design, and provide suggestions for activities to scaffold this process.

2 RELATED WORK

We situate our work within the broader paradigm of Assets-based design in CSCW and HCI [5, 8, 40], which draws from the paradigm of Assets-based community development (ABCD) [30]. Assets-based design aims to shift design from a deficit-based view to one leveraging communities' strengths, capacities, and assets (e.g., skills, relationships, and infrastructure). Our approach extends this paradigm by focusing in particular on how assets are linked to personal objects.

Some common co-design techniques foregrounding materials include Cultural probes [19], Bags of stuff [12], Toolkits [27, 39], and Contestational artifacts [10, 33]. Importantly, facilitators typically decide the materials in these techniques. Our approach differs as it aims to shift control of materials to users, thus enabling users to have more say over design.

Other techniques encourage users to bring in "things" from their lives, such as different types of End-user photography [35] including Photo-elicitation [18, 32], Photovoice [23], and Photo collaging [28], as well as Journaling [15, 36]. To support users to *share what they want in the format they want*, we extend the range of objects users can bring in and design with. This may include objects that are found, personally meaningful (e.g., mementos), tangible, intangible,



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analog, or digital. This way, our approach may also be used in more contexts and more frequently.

Two previous approaches focus on material-based prototyping using physical, locally-embedded objects [22]: Small change [21] focuses on “real” objects as prototypes drawing on and implemented into local practice, such as a bus stop prototype that engenders a marketplace and activates a local community [22]. A related, broader approach is Frugal innovation, which draws on renewable materials that are “*affordable, sustainable, lightweight, and rugged*” [2]. Our approach differs in that it is not focused specifically on prototyping, and it promotes contributions at the individual level.

Our approach aims to bridge the gap between previous approaches by 1) shifting control of design materials to users, 2) extending the range of objects users can bring in and design with, 3) promoting individuals’ contributions, and 4) foregrounding the potential of personal objects as design materials. From an empirical perspective, we also extend past work [9, 11, 13, 17, 26, 29, 37] by conducting empirical investigations specifically focusing on understanding the impact of personal objects on the design process and partners’ participation as designers.

3 INTEGRATING PERSONAL OBJECTS INTO CO- DESIGN WITH THE “4 C’S” APPROACH

Design Activities: To scaffold co-design and support participants’ bringing in and engaging with their objects, the research team suggested pre-workshop and in-workshop design activities. These activities had been iteratively developed and tested by the research team in pilots. They were created with the intention of integrating into communities’ existing practice, as well as being self-directed (i.e., without supervision from the research team). We structured design activities to build on one another so that participants’ objects were interwoven throughout the process. We were intentional in our design of the sequence of activities, but also wanted to allow for the flexibility of communities to pick and choose which activities to incorporate, when, how often, and for how long. Depending on the goals of the community, it may be helpful to repeat or omit certain design activities. At a high level, the design activities include (using the “4 C’s” mnemonic):

- **Choose:** Participants select object(s) to bring in before session (see prompt below); during the session, participants go around and share what they brought in and why.
- **Connect:** Participants draw connections between the objects they shared.
- **Consider:** Participants think of key considerations when creating solutions to their topic.
- **Create:** Participants brainstorm or design solutions to their topic.

Prompt: In addition to the suggested design activities, the research team iteratively developed a prompt that lends itself to the design topic at hand and supports participants to choose the personal objects to bring in. The prompt enabled participants to think about and engage with the project outside of workshop time [11], as well as encouraged forethought, reminded participants to bring in their objects, and was an opportunity to articulate what it was they were bringing in before sharing with the group. The workshop facilitator, who we refer to as Noora, and the lead author came up with

a generative prompt reflecting the theme of “Safe Spaces,” which was to “bring in something from your life that represents safety or makes you feel safe.” The young adults texted Noora what they planned to bring in a day before the workshop. We provide a template for prompts in Appendix A.1, which can be altered depending on the goals of the community.

4 CASE STUDY: METHODS

Partners: We collaborated with a nonprofit in Chicago focused on providing meaningful learning experiences for local youth underrepresented in STEM. A member of our research team, who had established trust through sustained engagement, facilitated our connection to the organization. Through a 1-year grant from Chicago, our partner organization hired 9 Black young adults (ages 16-21) as community ambassadors to organize safe spaces and programs for peers, with an emphasis on STEM. Noora, who is also a member of our partner organization’s leadership, led this effort. Noora and the young adults knew each other, had engaged in similar design activities, and could compare their experiences in this workshop with previous workshops. Table 1 more information on participants. We received informed consent from the young adults or their guardians and all aspects of this study were approved by our university’s review board.

Setting: The goal of the co-design workshop was for young adults to plan a “Safe Spaces” kickback event for peers, which entailed brainstorming event themes and activities. We were interested in how the young adults’ personal objects would help them achieve this goal. Noora and the lead author worked closely to embed our approach into existing processes and settings the young adults were familiar with. For example, design activities were structured like a discussion while Noora took notes on a whiteboard, as this was what the young adults were accustomed to. The co-design workshop also took place in a STEM learning center in a community space rented by the organization, in which all young adults had been to on numerous occasions. The co-design workshop lasted 1.5 hours. All co-design activities were conducted as a group.

Data Collection and Analysis: We triangulated data from a variety of sources to gain a better understanding of how young adults engaged with their materials in the design of a “Safe Spaces” event for peers. We collected observational and artifact data from the co-design workshop. To gather feedback at the end of the workshop, the young adults participated in a 15-minute focus group and completed individual surveys on their phones, which included short answer and slider questions. Examples of survey questions include: *What made you feel included/not included? If you have participated in similar sessions before, how would you compare this session to others you have participated in?* We also conducted 1-hour semi-structured interviews before and after the workshop with Noora. Recordings from the workshop and interviews were transcribed. To make sense of the data, we performed thematic analysis [3]. The lead researcher open coded the data to identify emerging themes. The research team then discussed and iterated upon emergent themes.

5 CASE STUDY: FINDINGS

How did Noora and the young adults instantiate the approach in their co-design workshop? **Choose:** Seated around a conference table (Figure 1), the young adults went around the room sharing their names, items they brought in that made them feel safe, and something they learned about another participant. Participants' objects and their descriptions are noted in Table 1 and listed in Figure 2. **Connect:** Noora then asked the young adults, "So in looking at the board and looking around at the people who have these kind of items, which do you feel connect or kind of like are on the same page, as far as what they do for somebody? Or any themes that you may notice?"

Noora drew arrows between the objects that participants made connections between. Participants drew the following connections: (1) Leire connected the crystal and fringes for spirituality; (2) Ana connected the necklace and headphones for overcoming difficulties; (3) Aisha connected the dog and Mr. Snuggles for comfort; (4) Rosa connected the crystal and necklace for how similar they are; (5) Emad connected the headphones and art for expressing emotions; (6) Ana connected the crystal and water bottle for radiating energy; and, (7) Zuri and Aisha connected the necklace and candle for serving as reminders. While participants drew connections between some objects more than others (i.e., Leire's crystal and Ana's necklace), participants found connections between every object at least once. **Consider:** Next, Noora posed the question, "Keeping in mind these items and what y'all discussed already today, what are some key things that you think we should consider when designing a 'Safe Space?'" Noora listed these considerations to the right of the whiteboard (Figure 2). Young adults mentioned the following 8 things to consider: (1) Good energy; (2) Boundaries; (3) Transparency; (4) Open-mindedness; (5) Respect of the space; (6) One mic, so one person talking at a time; (7) Correlation and how people connect; and (8) Love. It was evident that the previous two activities, Choose and Connect, helped inspire some key considerations like Good energy, Transparency, and Connection. **Create:** Lastly, the young adults brainstormed themes and activities for the spring kickback event (Figure 3). Young adults came up with themes including "Spring Fling," rebirth, and Hawaiian luau. Noora asked, "Do you all want to do any activities based off of the items that made y'all feel safe?" Leire, Zuri, and Ana suggested doing something relating to art (inspired by Eabha's object), such as having an art gallery or painting activity. The young adults agreed music was a given, harkening back to Aisha's headphones. Ana also suggested including affirmations inside Easter eggs, with statements like "You're so beautiful," an idea inspired by their objects that made them feel good.



Figure 1: Young adults seated around table;

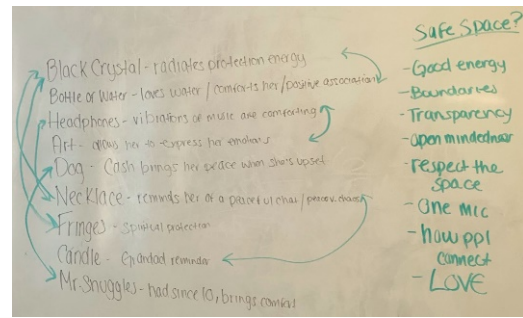


Figure 2: Choose, Connect, and Consider activities;

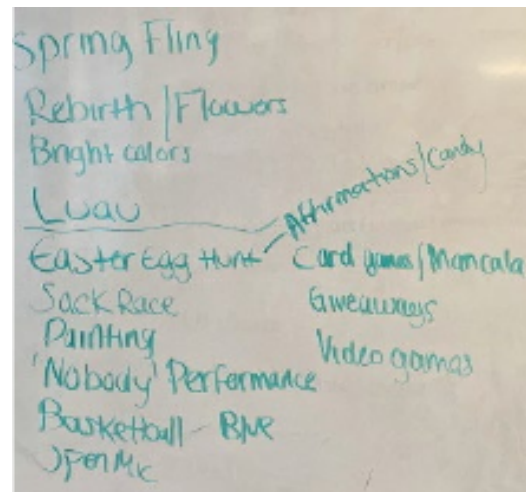


Figure 3: Create activity

How did the personal objects and our approach impact the design work? The young adults drew on their personal objects as sources of inspiration in the design of a kickback event for peers. We can trace how participants' objects, in Noora's words, "informed their designs." For example, Leire's crystal and Zuri's water bottle motivated the young adults to think about the importance of transparency and good energy in their event. Eabha's art inspired her peers to come up with art-related activities. Likewise, Ana came up with Easter egg hunt affirmations based on a peers' object that made them feel good. In the Noora's post-workshop interview, she said "like with the Easter egg hunt with the affirmations and all that kind of stuff I feel like directly came from what they talked about what made them feel safe versus usually it's just 'I saw a silent party last week that looked cool, let's do that.' You know, so definitely different." Along with drawing inspiration from their personal objects, the young adults felt their objects helped structure the conversation and co-design workshop. Sharing their personal objects was a way to get "everybody to talk, like it got me wanting to talk. . . It didn't feel like work, you feel me? You tricked us into working," as Leire expressed. Compared to previous workshops, participants thought this workshop was more structured. For example, Baoyi shared "This one was pretty quick, others usually drag too long, this one

Table 1: Participants' name (pseudonyms for anonymity), age, gender, race, and personal object (as described by participants).

Pseudonym	Age	Gender	Race	Personal Object	Object Description (In participants' words)
Leire	21	Female	Black	Black tourmaline crystal	"My crystal makes me feel safe because it radiates this protective energy. I usually just leave it in my car like it's all the bad juju bounces off of my car."
Zuri	18	Non-binary	Black	Water bottle	"The thing I brought today, or like my safe item, is a bottle of water. It makes me feel safe because I relate to water so much like our body is made up of 72% water. My sign is Aquarius, we are bearers of water. I just love water so much and water loves me."
Aisha	16	Female	Black	Headphones	"My safe thing that I brought was my headphones because I love music. Like the vibrations from music exude me. It's like when I turn it up on the highest level I can't hear nothing else around me."
Eabha	17	Female	Black	Art [pictures on phone]	"The thing I brought in was my art, and I say art because it allows me to get away from my thoughts."
Coral	18	Female	Black	Dog [pictures on phone]	"I brought a picture of my dog [named Cash] because he is really my baby. He is so sweet, even though he can get his moments when I'm gone."
Ana	18	Female	Black	Necklace	"The thing that I brought in today is a necklace. It represents cause and effect within my life. It's like chaos to peace. Before there was chaos and then I feel like I connected with a talk that brought peace in my life [necklace reminds her of a talk]."
Emad	17	Male	Black	Fringes on clothes	"I brought my fringes that I wear every day [unzips jacket to show shirt underneath]. These are something I wear every day because I follow the Bible. These are something that God told us to wear to protect us and to keep us with like a righteous mind."
Baoyi	17	Male	Black	Candle	"I brought a candle from my grandpa who passed away. He used to make me feel safe."
Rosa	16	Female	Black	Mr. Snuggles	"I brought in Mr. Snuggles [stuffed animal in the shape of a purple sloth]. He makes me feel happy and I've had him since I was 10 years old."

was perfect." Ana added, "It was good, it was like very accordant, like everything was in order." Noora also thought this workshop was more intentional and thoughtful: "I will say it was a lot more intentional of a session. because a lot of it [previous sessions] is just kind of them thinking in the air. We talk, we brainstorm, and then we vote on the best ideas and like, although that still solicits results, they're less thoughtful of a result." Noora considered what the session would have looked like had the young adults only participated in the first design activity (Choose): "I think the process was helpful. I think, just bringing in items minus the process probably would not have elicited the same feedback."

How did the personal objects and our approach impact the young adults as designers? Overall, the young adults shared that they felt more included in the co-design process. Sharing their personal objects was a way for young adults to open up to their peers. Ana recounted how "it made me feel good to have a nice, safe place to open up to tell people about my experience, my cause and my effect." Similarly, Emad felt "It was good, got to show who I am to my coworkers." Young adults recounted how the experience of bringing in and sharing their personal objects helped them to feel heard and seen. In their post-workshop survey responses, the young adults shared what made them feel included: (1) Zuri wrote "people listening to me, I don't feel like many people do that." (2) Baoyi wrote,

"how everyone spoke to each other and made you feel included." (3) and Emad wrote, "I had a voice." Only one participant, Zuri, shared in the focus group and survey that "it was scary, I don't like letting people in my personal life," but liked the process of "getting to know and understand each other better." Zuri's feedback was an important reminder of the need for mechanisms that support different social preferences or personalities in co-design. The personal objects also led to outcomes participants could carry over into other workshops and their personal lives. For example, personal objects were a way of building connections and fostering common ground. In Noora's words, "I think finding the connections between the artifacts helped the students to find connections just between each other in general and find common ground." The young adults recounted how they enjoyed connecting with their peers and how opening up to one another made connecting easier. Leire stated, "It made me feel more connected and like, I feel like when people open up you can actually like connect with people. You get to do some introspection with people you've been working with." Additionally, young adults emerged with a newfound understanding of their objects and what can constitute a design material. For example, Ana shared that "It made me want to think more about things on a deeper level, like if I got a teddy bear at home I can take this thing home and do it to my teddy bears."

6 DISCUSSION

Previous work has documented instances where Participatory design has fallen short of its inclusive ideals [14, 24, 25]. One aspect of Participatory design, materials, is critical to the experience and participation of users—as well as design output—yet is often overlooked or imposed on users. In this paper, we proposed an approach that shifts control of design materials to users by inviting them to bring in personal objects from their lives. Extending prior work on Assets-based design [5, 8, 40], we aimed to leverage tangible or intangible artifacts that participants choose to bring in. To support participants in their selection and use of their objects, we suggested a set of pre-workshop and in-workshop design activities (the “4 C’s”) as well as a prompt. We presented observations and direct accounts from young adults and the workshop facilitator into their experiences of engaging in the co-design of a “Safe Spaces” kick-back event for peers. From these observations and accounts, we see how the young adults were able to share aspects of their identities and lived experiences in the design process, perhaps more than in previous workshops. The outcomes of Participatory design engagements may not always be tangible solutions, but may be increases in self-efficacy, capacities, and shifts in mindsets. An example of this is how Hussain et al. focused on building local capacity of Cambodian prosthetists and engineers by teaching them design principles so that future design projects could be carried out independent of researchers [26]. In this case study, we encouraged the young adults to think about their objects in new ways and to recognize the viability of everyday objects, whether tangible or intangible, as design material. An outcome of this was participants, like Ana, wanting to adopt this approach outside of the immediate design context and on their own. In future work, we will explore the scalability of this approach and how other groups embed our approach in their co-design workshops. We are curious how this approach can be adapted across design contexts, modalities (e.g., Distributed Participatory design [6, 16]), and beyond a single workshop; what happens when the design activities are implemented differently than suggested; what happens when participants do not know each other; and how the personal objects can be used in physical prototyping of co-design solutions.

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A APPENDICES

A.1 Prompt Template

“To help us with planning and to make the session more personal, we’re asking everyone to bring in something from your life that represents safety or makes you feel safe. You will be able to share

PROMPT

this with the group. This can be a physical object like a handmade item or book, or it can be digital like a photo or audio recording, or something else we haven’t thought about! We’re not looking for anything in particular, so feel free to be as creative as you wish, and you can bring in more than one thing to share! We’ll be doing a “show and tell” and then using whatever you all bring in for inspiration and brainstorming.

IMPORTANT: By _____, please text/email

DATE

_____ the thing(s) that you will be sharing with

FACILITATOR

the group. Please include a 1-2 sentence description, photo, or link.”